

Pengyu Zhang

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• ABOUT ME

I work on machine learning for knowledge graphs, where the data changes over time and many entities are hard to tell apart. I build models that use graph evolution, user activity and time itself to keep entity linking, knowledge graph completion and recommendation accurate on sparse and long-tail entities. The same problem appears in industry as entity resolution under data drift, for example products, authors, places or merchants whose records look alike and keep changing. My first-author methods outperform the previous state of the art by up to **20.9%** in entity linking three years after training, and by up to **58.2%** on the most ambiguous 1% of entities. Code and benchmarks are on GitHub.

Open to Applied Scientist, Machine Learning Engineer and Data Scientist roles, and postdoc positions.

Multi-modal knowledge graphs

Entity linking & resolution

Link prediction

Recommendation

LLMs

• EDUCATION

2022 – present **PhD in Computer Science**, University of Amsterdam. Defense planned for March 2027.

Thesis: *Addressing Temporal Dynamics and Entity Ambiguity in Knowledge Graph Tasks with Multi-Source Information*.

Promotor [Prof. Dr. Paul Groth](#); co-promotor [Dr. Klim Zaporozhets](#).

2019 – 2022 **MEng in Control Engineering**, Beijing University of Technology. Supervisor [Prof. Dr. Yong Zhang](#). Outstanding Master's Thesis award.

• EXPERIENCE

PhD Candidate · INDElab, University of Amsterdam

2022 – present

Five first-author projects, with code and data released for every published one. Largest settings: graphs of up to 5.4M entities and 2M+ training samples, and entity linking over 21 million tweets.

CYCLE & TIGER Temporally robust entity linking

CIKM 2024 & ECAI 2024

- Entity linkers are trained on one snapshot of the knowledge graph but used for years while the graph keeps changing. I designed TIGER, which adds graph structure to a text-only bi-encoder, and CYCLE, which turns relations that appear or vanish between yearly snapshots into contrastive positives and negatives.
- Against the 2023 state of the art, at a three-year gap, TIGER gains **20.9%** on continual entities and CYCLE gains **17.8%** averaged over continual and new entities, most on low-degree ones. Both ship with public benchmarks (Graph-TempEL, GCL-TempEL).

Time Imprint Time as a modality for look-alike entities

under review, arXiv 2026

- Some entities share almost identical text and images, like Napoleon Bonaparte and the 2023 film Napoleon. I treat time as an entity-level modality and inject an attention-pooled time embedding into the encoder, the scoring function and a contrastive objective.
- New state of the art on MKG-W with Hits@1 up **6.1%**. On the most ambiguous 1% of entities, Hits@1 improves by up to **58.2%**.

Beyond Images Data-centric enrichment of multi-modal knowledge graphs

ESWC 2026

- Curated image sets drop logos and symbols because visual encoders turn them into noise. I built a pipeline that retrieves more than 300,000 Wikipedia images, captions them with vision-language models and fuses each entity's captions into one text summary with an LLM, with no model changes.
- Hits@1 rises **4.8% to 7.8%** across three datasets and four models; on a hand-picked set of 20 logo and symbol entities it more than quadruples. Three enriched datasets released.

TimeRoute Time-aware multi-modal recommendation

under review, arXiv 2026

- Recommenders usually fuse text, images and audio with one global weighting, but which modality matters changes over time and differs between users. I built a per-user router that turns interaction timestamps into modality weights, and a time-conditioned diffusion module that filters outdated edges before graph propagation.
- Best on every metric on TikTok, Amazon-Baby and Amazon-Sports over ten seeds, **4.9% to 9.8%** above the strongest baselines. Replacing the timing signal with noise removes the gain.

Select, Don't Train Modular entity disambiguation with LLM selection

ISWC 2026

- Dual-encoders handle retrieval and fine-grained selection in one embedding space and need retraining when the knowledge base changes. With Fina Polat (lead), we split the two: a retriever returns 16 candidates and an LLM picks one or abstains.
- Plain BM25 with no retriever training reaches **86.3 micro-F1** on ZELDA, 4 points above the best trained dual-encoder; with a trained dense retriever it reaches 88.5.

Graph-TempCZ Large-scale temporal link prediction

LREC 2026

- Software-mention data had no graph structure or time axis. With Congfeng Cao (lead), we built a publication-software graph from 1.5 million biomedical papers, with more than 6.5 million edges from 1959 to 2022, and framed software usage as temporal link prediction.
- GraphSAGE reaches **92.9%** accuracy, 6 points above the best feature-based baseline. Accuracy decays as the gap between training and test years grows, with a roughly two-year transfer window.

Fusion Training Data mixing for hybrid-reasoning LLMs

ACL 2026 SRW

- Hybrid models like Qwen3 can answer quickly or reason step by step, but the two kinds of training data interfere. With Congfeng Cao (lead), we swept seven data ratios and three training schedules on Qwen3-4B and released the Fusion Bench benchmark.
- More short-answer data steadily lowers long-form reasoning accuracy, the two accuracies are negatively correlated, and interleaving the two is the most robust schedule.

MEng Student · Beijing University of Technology, group of Prof. Dr. Yong Zhang

2019 – 2022

Visual Analysis Interactive author name disambiguation system

JCAD 2022

- Research offices separate authors who share a name by hand, and a bare classifier gives them no way to check its decisions. I built and deployed a visual analytics system that pairs a multi-view graph classifier (about **91%** accuracy on ACM and DBLP) with linked views of collaboration structure and raw metadata.
- On 4,000 real university papers, nine graduate students and two experienced administrators completed the disambiguation tasks with over **95%** success; usability scored 84 to 88 out of 100.

DualChannel Text-plus-graph author name disambiguation

Information 2021

- With Xin Zheng (lead), we combined text semantics with a co-author, organisation and venue graph to separate same-name authors over 205,000 papers on AMiner WholsWho.

• SKILLS

Programming	Python, SQL (PostgreSQL), Bash, Git, Linux.
ML tools	PyTorch, Hugging Face, scikit-learn, NumPy, Pandas, Weights & Biases; multi-GPU training on A100 and H100.
Methods	Graph neural networks, contrastive learning, multi-modal fusion, diffusion models, temporal representation learning, entity linking and resolution, recommendation, LLM and vision-language pipelines.
Languages	Chinese (native), English (fluent, IELTS 7.0), Dutch (elementary, A2).

• AWARDS & SERVICE

Awards	Conference travel grant, ACL 2026. Outstanding Master's Thesis, Beijing University of Technology, 2022.
Talks	Paper talks at ACL, ESWC and LREC 2026; EKAU, CIKM and ECAI 2024.
Service	Reviewer or sub-reviewer for CIKM, ECAI, ESWC and PKDD.
Teaching	Teaching assistant, Data Engineering, Beijing University of Technology, 2020 – 2021.

• PUBLICATIONS

- [14] TimeRoute: Time-Aware Modality Routing and Diffusion for Multi-Modal Recommendation. **Pengyu Zhang**, Yangqin Jiang, Klim Zaporozets, Congfeng Cao, Paul Groth. Under review. [arXiv](#)
- [13] Time Imprint: Learning Time-Aware Representations in Multi-Modal Knowledge Graphs. **Pengyu Zhang**, Klim Zaporozets, Congfeng Cao, Jia-Hong Huang, Paul Groth. Under review. [arXiv](#)
- [12] Select, Don't Train: The Benefits of Modular Entity Disambiguation with LLM-Based Selection. Fina Polat, Daniel Daza, **Pengyu Zhang**, Klim Zaporozets, Paul Groth. *ISWC 2026*. [arXiv](#) · [Code](#)
- [11] Fusion Training for Mathematical Generalization in Large Language Models. Congfeng Cao, **Pengyu Zhang**, Jelke Bloem. *ACL 2026, Student Research Workshop*. [Paper](#) · [DOI](#) · [Code](#)
- [10] Are a Thousand Words Better Than a Single Picture? Beyond Images – A Framework for Multi-Modal Knowledge Graph Dataset Enrichment. **Pengyu Zhang**, Klim Zaporozets, Jie Liu, Jia-Hong Huang, Paul Groth. *ESWC 2026*. [Paper](#) · [DOI](#) · [Code](#) · [Demo](#) · [YouTube](#) · [Bilibili](#)
- [9] Graph-TempCZ: A Graph Representation of Software Mentions for Predicting Software Usage in Scientific Publications. Congfeng Cao, **Pengyu Zhang**, Jelke Bloem. *LREC 2026*. [Paper](#) · [DOI](#) · [Code](#)
- [8] A Survey of Large Language Models for Data Challenges in Graphs. Mengran Li, **Pengyu Zhang**, Wenbin Xing, Yijia Zheng, Klim Zaporozets, Junzhou Chen, Ronghui Zhang, Yong Zhang, Siyuan Gong, Jia Hu, Xiaolei Ma, Zhiyuan Liu, Paul Groth, Marcel Worring. *Expert Systems with Applications 2026*. [Paper](#) · [DOI](#) · [Code](#)
- [7] Understanding the Impact of Entity Linking on the Topology of Entity Co-occurrence Networks for Social Media Analysis. James Nevin, **Pengyu Zhang**, Dimitar Dimitrov, Michael Lees, Paul Groth, Stefan Dietze. *EKAU 2024*. [Paper](#) · [DOI](#) · [Code](#)
- [6] CYCLE: Cross-Year Contrastive Learning in Entity-Linking. **Pengyu Zhang**, Congfeng Cao, Klim Zaporozets, Paul Groth. *CIKM 2024*. [Paper](#) · [DOI](#) · [Code](#)
- [5] TIGER: Temporally Improved Graph Entity Linker. **Pengyu Zhang**, Congfeng Cao, Paul Groth. *ECAI 2024*. [Paper](#) · [DOI](#) · [Code](#)
- [4] Relationship Updating Network with Contrastive Learning. **Pengyu Zhang**, Yong Zhang, Xinglin Piao, Yongliang Sun, Baocai Yin. *Physica A 2024*. [Paper](#) · [DOI](#) · [Code](#)
- [3] MVMA-GCN: Multi-view Multi-layer Attention Graph Convolutional Networks. **Pengyu Zhang**, Yong Zhang, Jingcheng Wang, Baocai Yin. *Engineering Applications of Artificial Intelligence 2023*. [Paper](#) · [DOI](#) · [Code](#)
- [2] Visual Analysis for Name Disambiguation of Academic Papers (in Chinese). **Pengyu Zhang**, Yong Zhang, Yanjie Cui, Baocai Yin. *Journal of Computer-Aided Design & Computer Graphics 2022*. [Paper](#) · [DOI](#) · [Code](#) · [Demo](#) · [YouTube](#) · [Bilibili](#)
- [1] Dual-Channel Heterogeneous Graph Network for Author Name Disambiguation. Xin Zheng, **Pengyu Zhang**, Yanjie Cui, Rong Du, Yong Zhang. *Information 2021*. [Paper](#) · [DOI](#) · [Code](#)